What Is an Extra 100 Square Feet Worth?
Living Area and Sale Prices in Ames, Iowa, 2006–2010

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Repository: https://github.com/shanshiny06/econ3916-midterm

Notebook: midterm-starter.ipynb - Colab

Question	Among residential home sales in Ames, Iowa, 2006–2010, how much higher is the sale price (US$) of a home with 100 more square feet of above-ground living area, holding its neighborhood, overall quality, year built and year of sale fixed?
Headline finding	100 more square feet of living area goes with a sale price about 2.75% higher (robust 95% CI 2.16% to 3.35%, N = 1,460), roughly US$4,490 for a median-priced home. This is a conditional association, not a causal effect: larger homes come with features the model does not hold fixed. Without the four homes above 4,000 sq ft, the estimate is 3.18%.
Data	One row is one residential sale (N = 1,460). Ames City Assessor's Office records compiled by De Cock (2011); the Kaggle "House Prices" training set, downloaded from OpenML (dataset 42165) on 2026-11-05.
Method	OLS of log sale price on living area (in units of 100 sq ft), the assessor's overall quality rating, year built, and neighborhood and year-of-sale dummies, with heteroskedasticity-robust (HC1) standard errors.

1. The Question and the Data

Floor area is the first thing buyers compare when they look at a home, and one of the first things a city assessor records when valuing it for property tax. How much an extra 100 square feet adds to a sale price matters to buyers and sellers deciding what a larger home is worth, to owners weighing an extension, and to assessors, whose valuations rest on exactly this kind of comparison. This report asks: among residential home sales in Ames, Iowa between 2006 and 2010, how much higher is the sale price of a home with 100 more square feet of above-ground living area, holding its neighborhood, overall quality, year built and year of sale fixed? The outcome is the sale price in US dollars, modelled in logs; the main explanatory variable is above-ground living area, measured in hundreds of square feet.

The data are a product of an assessor's measurement decisions, not a survey designed for this question. The Ames City Assessor's Office records each sale in order to value property for tax. De Cock (2011) started from the office's file of 3,970 sales described by 113 variables, removed variables tied to weighting and adjustment factors in the city's own valuation model, and published 2,930 residential sales as a teaching dataset. The file used here is the Kaggle "House Prices: Advanced Regression Techniques" training set, the 1,460 of those sales released with their prices, downloaded from OpenML

(dataset 42165) on 2026-11-05. One row is one sale, and the file has 81 columns. Three measurement decisions shape the variables I use. Living area counts only above-ground space, so a finished basement adds nothing to it. The overall quality rating is an assessor's judgment of materials and finish on a 1–10 scale, not a measurement. And prices are recorded in current dollars of the year of sale, a difference the year-of-sale dummies absorb.

Missing values come in three kinds. Most are not missing data at all. PoolQC, Alley, Fence, FireplaceQu and the garage and basement variables are coded NA when the home has no such feature, and cross-tabulations confirm it exactly: all 1,453 missing pool-quality values belong to homes with no pool area, and all 81 missing garage types to homes with no garage area. These gaps are a coding decision, not lost information. Lot frontage, the length of the lot that faces the street, is genuinely missing for 259 sales (17.7%). The gap depends on an observed variable: 52.1% of cul-de-sac lots lack a value, against 12.7% of inside lots, plausibly because frontage is hard to define on an irregular lot. That makes MAR the defensible label, although MNAR cannot be ruled out from the data alone. The electrical system (1 sale) and masonry veneer (8 sales) are missing too rarely to show any pattern and are treated as MCAR. None of these variables enters the model, and none of the model's variables has a missing value, so the regression uses all 1,460 sales.

The missingness that matters for the answer is the homes that are not in the file. Only homes that sold appear; homes that were never listed, or listed and did not sell, are absent, and whether a home sells plausibly depends on its asking price relative to its value. In that sense the gap is MNAR, and the results describe the market for homes that sold in Ames between 2006 and 2010, a period that includes the 2008 housing downturn. They may not carry over to the full housing stock or to other cities. Table 1 summarises the variables the model uses.

Table 1. Summary statistics, 1,460 home sales in Ames, Iowa, 2006–2010

Variable (units)	Mean	Median	SD	Min	Max	N
Sale price (US$)	180,921	163,000	79,443	34,900	755,000	1,460
Log sale price	12.02	12.00	0.40	10.46	13.53	1,460
Above-ground living area (sq ft)	1,515	1,464	525	334	5,642	1,460
Overall quality rating (1–10)	6.10	6	1.38	1	10	1,460
Year built	1971.3	1973	30.2	1872	2010	1,460
Year sold	2007.8	2008	1.33	2006	2010	1,460
Rooms above ground	6.52	6	1.63	2	14	1,460
Bedrooms above ground	2.87	3	0.82	0	8	1,460

Sale prices in current US dollars of the year of sale. The 1,460 sales cover 25 neighborhoods, which enter the model as dummy variables. Source: Ames City Assessor's Office via De Cock (2011), Kaggle training set, downloaded from OpenML (dataset 42165) on 2026-11-05.

2. Descriptive Evidence

Sale prices are strongly skewed to the right (Figure 1). The mean price, US$180,921, sits well above the median, US$163,000, because a long tail of expensive sales stretches up to US$755,000, while 77.3% of

sales fall between US$100,000 and US$250,000. The skewness is 1.88. In logs the distribution is nearly symmetric (skewness 0.12), so the model uses log sale price, and its coefficients read as approximate percentage differences.

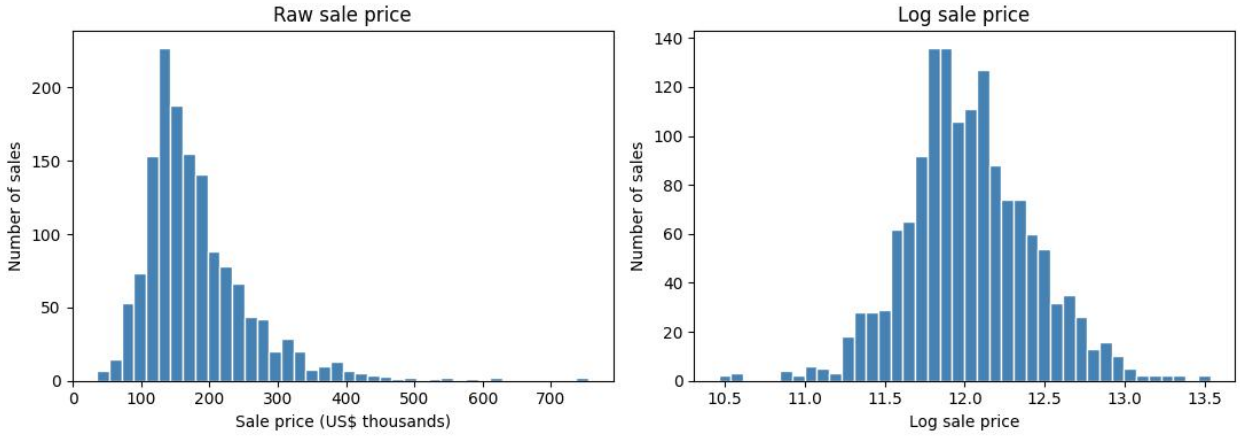


Figure 1. Sale prices are strongly right-skewed (skewness 1.88): 77.3% of homes sold for US$100,000–250,000, with a long tail up to US$755,000; in logs the distribution is nearly symmetric (skewness 0.12). Source: Ames housing data, De Cock (2011), via OpenML.

Figure 2 shows the relationship this report is about. Log sale price rises steadily with above-ground living area, with a correlation of 0.70, and the cloud of points is close to a straight line. Four sales exceed 4,000 square feet. Two of them sit near the trend; the other two sold for far less than their size predicts, and both were partial sales, recorded before a newly built home was finished. Section 4 returns to them.

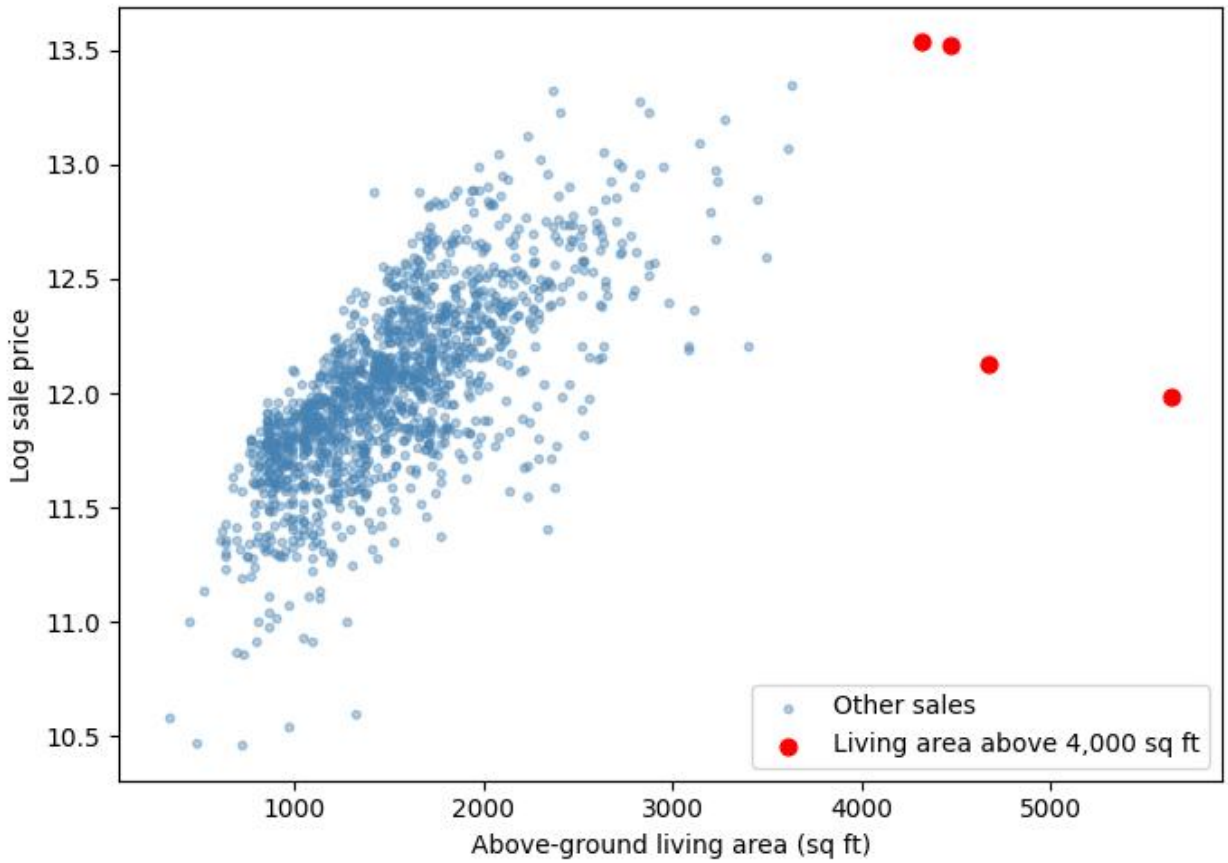


Figure 2. Larger homes sell for more: log sale price rises steadily with living area (correlation 0.70); two of the four homes above 4,000 sq ft (red) sold far below the trend, both as partial sales of unfinished homes. Source: as Figure 1.

Figure 3 is the reason the raw relationship cannot be read on its own. Location moves prices a great deal: the median sale price runs from US$88,000 in Meadow Village to US$315,000 in Northridge Heights, 3.58 times as much. If larger homes are concentrated in the expensive neighborhoods, part of the slope in Figure 2 is location rather than size, which is why the model holds neighborhood fixed.

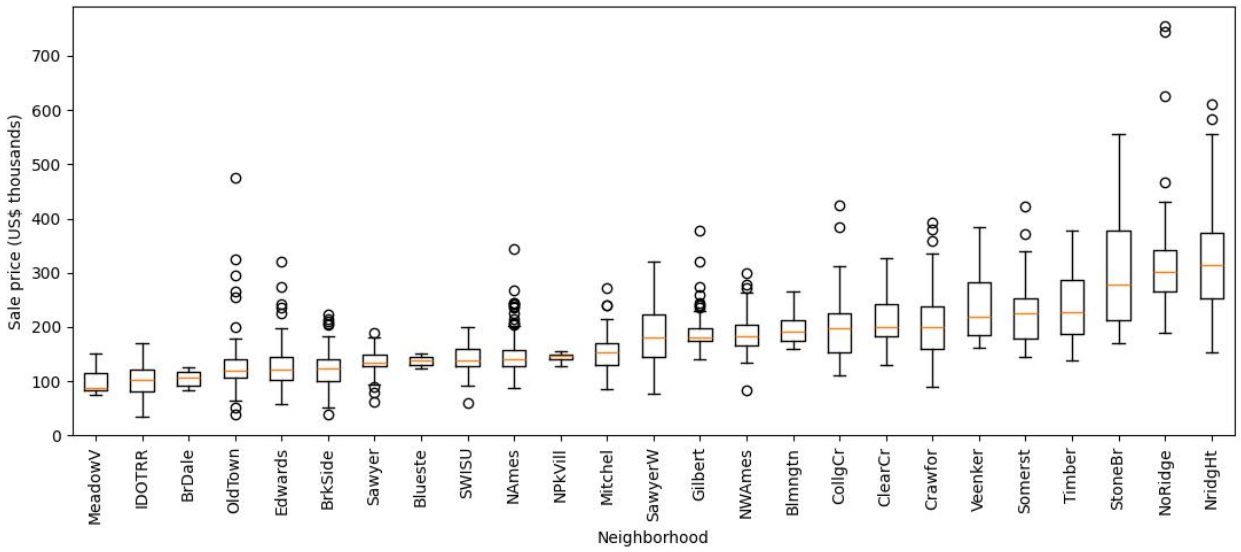


Figure 3. Location alone moves prices substantially: median sale prices range from US$88,000 in Meadow Village to US$315,000 in Northridge Heights, a 3.58-fold difference. Neighborhoods are ordered by median price; the price axis starts at zero. Source: as Figure 1.

3. The Model

I fit the following model by OLS, with heteroskedasticity-robust (HC1) standard errors, for the reason given in Section 4:

$$log(Price_i) = \beta_0 + \beta_1\, Area_i + \beta_2\, Quality_i + \beta_3\, YearBuilt_i + Neighborhood_{n(i)} + YearSold_{t(i)} + u_i$$

Area is above-ground living area in hundreds of square feet, Quality is the assessor's 1–10 rating, and the last two terms are sets of dummy variables for the neighborhood and the year of sale. Because the outcome is in logs, a coefficient b means the price changes by about 100 × (e^b − 1) percent for a one-unit change. Table A1 in the appendix reports three versions: living area alone (column 1), the main model (column 2), and the main model with room counts added (column 3).

Living area. Holding neighborhood, overall quality, year built and year of sale fixed, a home with 100 more square feet of above-ground living area sold for about 2.75% more. At the median price of US$163,000 that is roughly US$4,490. The estimate is precise: the robust 95% confidence interval runs from 2.16% to 3.35%, so the data rule out both a negligible effect and one as large as the 5.47% that living area alone suggests (column 1). The controls cut the raw association in half, which shows how much of the simple relationship in Figure 2 reflects location, quality and age rather than size. Together they raise the R^2 from 0.49 to 0.83.

Overall quality. Each additional point on the assessor's 10-point quality scale goes with a sale price 11.27% higher, about US$18,374 at the median price, holding size, location, age and year of sale fixed. The robust 95% confidence interval runs from 9.92% to 12.64%.

Room counts. Adding the number of rooms and bedrooms above ground barely moves the area effect, from 2.75% to 2.67% (column 3), and neither room count is statistically distinguishable from zero with robust standard errors. Little of the value of extra space works through the number of rooms. Each

control in the main model has a stated role:

- **Neighborhood dummies block a confounding path.** Location affects both how large homes are, since some areas were developed with larger lots and floor plans, and what they sell for, through schools, amenities and distance to campus. Without them the area coefficient would partly measure location. They would be a bad control only if a home's size determined its neighborhood, which it cannot.
- **Overall quality blocks a confounding path, with a caveat.** Builders who put in larger floor plans also tend to use better materials and finishes, and quality raises price whatever the size. But if assessors rate larger homes as higher quality simply because they are larger, quality sits partly on the path from area to price, and controlling for it pushes the area coefficient towards zero.
- **Year built blocks a confounding path.** Newer homes are built larger and also sell at a premium for newer systems and design. Size cannot change when a home was built, so year built cannot be a bad control.
- **Year-of-sale dummies mainly improve precision.** They absorb market-wide price swings, including the 2008 downturn, that hit every home sold in a given year. They would confound the area estimate only if the size of homes sold changed systematically from year to year.
- **Rooms and bedrooms are left out of the main model on purpose.** Adding floor area typically adds rooms, so room counts are partly caused by the explanatory variable and would be bad controls. Column 2 measures the full price difference that comes with extra space; column 3 measures the difference holding the number of rooms fixed, that is, of having larger rather than more rooms.

4. Diagnostics

Residuals. Figure A1 in the appendix plots the main model's residuals against the fitted values and against living area. They scatter evenly around zero, with no curve, so a straight line in living area is a reasonable shape for log price. Two sales stand far below every other point: the two largest homes, the partial sales from Figure 2, which sold for much less than the model predicts.

Constant variance. The Breusch–Pagan test rejects constant variance (statistic 202.90, $p < 0.001$). The coefficients are still unbiased, but the conventional standard errors are not valid, so every standard error and confidence interval in this report is heteroskedasticity-robust (HC1). The choice matters: the robust standard error on living area is almost three times the conventional one (0.0029 against 0.0011), which widens its 95% confidence interval to 2.16%–3.35% per 100 square feet. The conclusion that living area carries a sizeable premium does not depend on it.

Collinearity. The variance inflation factors are 1.87 for living area, 2.97 for overall quality and 4.92 for year built, all below the usual threshold of 5, so overlap among the regressors does not meaningfully inflate the uncertainty of the coefficients I interpret. Year built is the highest because Ames's neighborhoods were built up in different decades, so location partly predicts a home's age. The largest VIF among the dummies, 13.65 for North Ames, reflects overlap among the neighborhood dummies themselves, not with living area.

Influence. With the rule-of-thumb cutoff $4/n = 0.0027$, 92 sales exceed the threshold for Cook's

distance, but with 1,460 sales the cutoff is loose. Two sales stand out by an order of magnitude: the two partial sales above 4,000 square feet, with Cook's distances of 0.278 and 0.098, against 0.024 for the next sale. Re-estimating the main model without the four homes above 4,000 square feet, the cutoff De Cock (2011) himself suggests, raises the living-area effect from 2.75% to 3.18% per 100 square feet and cuts its robust standard error from 0.0029 to 0.0012. These few sales pull the estimate down and account for most of the gap between the robust and conventional standard errors. I keep all sales in the main results, since dropping observations because of how they fit is a choice a reader should see, and report that the effect lies between about 2.75% and 3.18% depending on whether these unusual transactions are included.

5. Threats and Conclusion

For the 2.75% estimate to be the causal effect of adding 100 square feet to a home, living area would have to be unrelated to every other determinant of price among homes in the same neighborhood, with the same quality rating, built in the same year and sold in the same year. That condition probably does not hold. First, the model omits features that tend to come with larger homes and also raise prices, such as lot size, basement area, garage size and the number of bathrooms. Because these are likely positively related to both living area and price, the estimate is more likely to overstate than understate the value of space. Some of them, such as bathrooms, are themselves partly caused by floor area, so adding them would raise the same bad-control problem as room counts. Second, the quality rating may partly reflect size if assessors rate larger homes more highly, which would push the estimate the other way, towards zero. Third, any random error in recorded living area would also bias the coefficient towards zero. Fourth, the sample contains only homes that sold, and two partial sales of unfinished homes are influential enough to move the estimate. Finally, the regression compares different homes rather than the same home before and after an extension, so it does not tell an owner directly what an addition would add to their home's value.

The estimate is therefore a conditional association: among homes in the same neighborhood, with the same quality rating, built in the same year and sold in the same year, homes with 100 more square feet of above-ground living area sold for about 2.75% more in Ames between 2006 and 2010. A stronger design would use repeat sales of the same homes and compare the change in price for homes that added living space with the change for homes that did not, a difference-in-differences. It would remove every fixed feature of each home and location, but it would assume that both groups' prices would have moved in parallel without the addition, and it would need standard errors this course has not covered, so I do not estimate it.

Conclusion. Larger homes sold for substantially more in Ames. Holding location, quality, age and year of sale fixed, 100 extra square feet of living area goes with a price about 2.75% higher (robust 95% CI 2.16% to 3.35%), roughly US$4,490 for a median-priced home. Comparing homes in the same neighborhood and of similar quality cuts the raw association in half, from 5.47%. The estimate barely changes when room counts are added and rises to 3.18% when the largest homes are excluded, so a reader can be confident that living area carries a sizeable premium in this market, but should treat the exact figure as an association rather than the value an extension would add. With another month I would add lot size and basement area as further controls, check whether the premium differs across

neighborhoods or price levels, and repeat the analysis on De Cock's full sample of 2,930 sales.

References

De Cock, D. (2011). Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project. *Journal of Statistics Education*, 19(3). https://doi.org/10.1080/10691898.2011.11889627

Kaggle. House Prices: Advanced Regression Techniques, training data. https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques

OpenML. house_prices, dataset 42165. https://www.openml.org/d/42165

Appendix

Table A1. Log sale price regressed on above-ground living area and controls

	(1) Area only	(2) Main	(3) Adds room counts
Living area (100 sq ft)	0.0533	0.0272	0.0264
	(0.0030)	(0.0029)	(0.0054)
Overall quality (1–10)		0.1068	0.1060
		(0.0062)	(0.0065)
Year built		0.0026	0.0026
		(0.0005)	(0.0005)
Rooms above ground			0.0092
			(0.0083)
Bedrooms above ground			−0.0167
			(0.0119)
Constant	11.2166	5.7815	5.8001
	(0.0439)	(1.0431)	(0.9439)
Neighborhood dummies	No	Yes	Yes
Year-of-sale dummies	No	Yes	Yes
N	1,460	1,460	1,460
R^2	0.491	0.828	0.829

Note: Dependent variable: log of the sale price in US dollars. OLS with heteroskedasticity-robust (HC1) standard errors in parentheses. Column 2 is the main model. Column 3 adds the numbers of rooms and bedrooms above ground, which living area partly determines, to show how much of the area effect runs through room counts (Section 3). Neighborhood and year-of-sale coefficients are not shown.

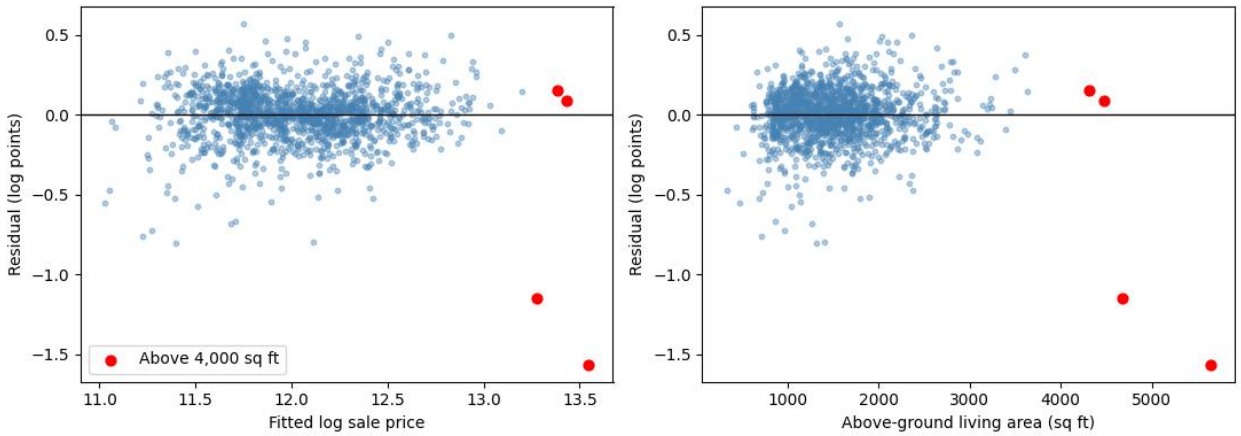


Figure A1. The main model's residuals show no curve against the fitted values (left) or living area (right); the two large negative outliers are partial sales of homes above 4,000 sq ft. Source: as Figure 1.

Reproducing the numbers. Every number in this report is printed by the last code cell of the notebook in the repository, "Numbers my report quotes". The README gives the data source and the order to run

the cells.

Use of AI. With the written permission of the course instructor, Dr. Piao, I used an AI assistant (Claude) to help write code, interpret results and draft text. I ran every step myself and can explain each line of the notebook.